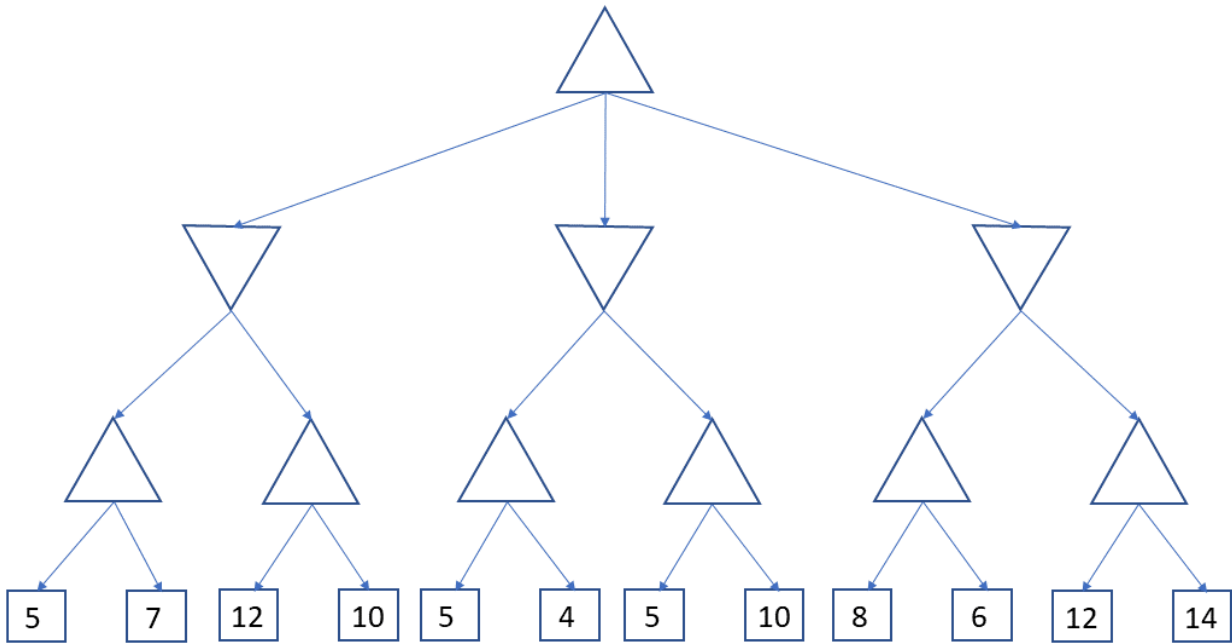
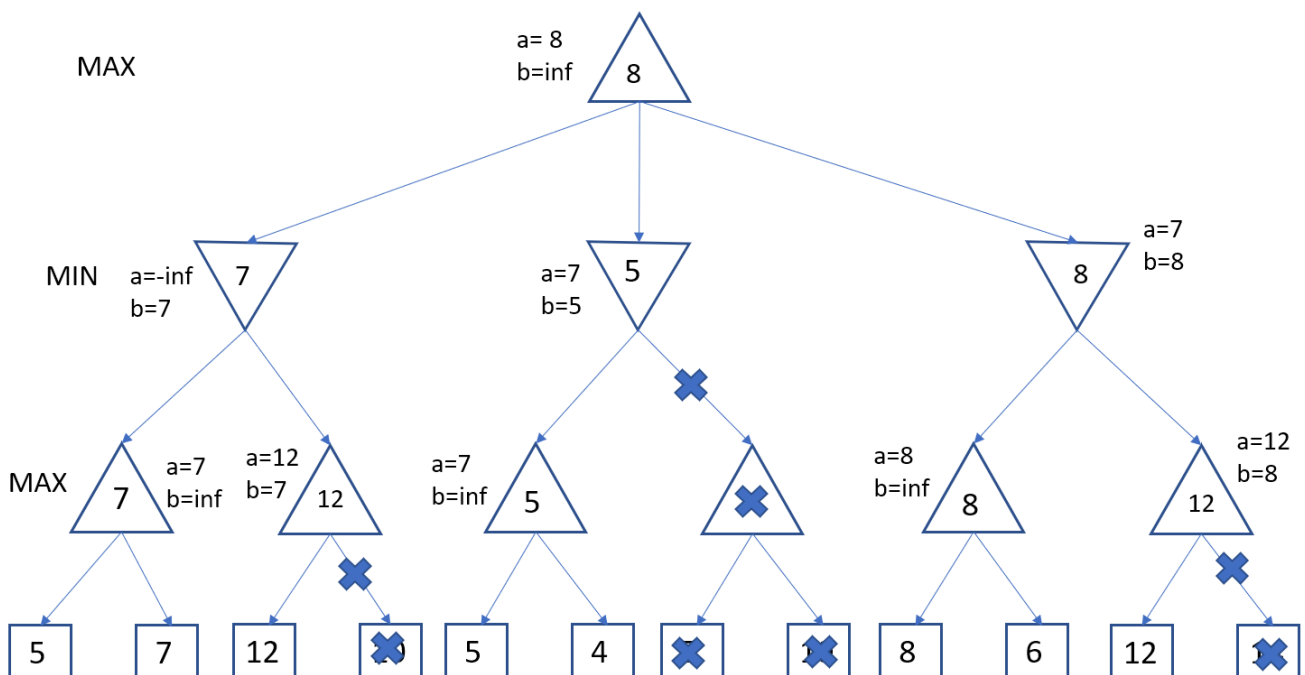


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1. Simulate **alpha-beta pruning** for the following game tree. For each node, write the minimax value inside the node, and write the values of alpha and beta beside the node. Also show which branches and nodes will be pruned by putting a cross mark on/beside them. [Do this ON the figure itself.] [7]



Solution:



2. An attendee must plan their conference day (9 AM – 6 PM) by arranging **five sessions** in an optimal order. Each session can be attended at most once.

Available sessions: Keynote, Workshop, Poster Viewing, Networking Lunch, Panel Discussion, Coffee Break

Session Constraints

A. Avoid (soft constraints)

- **Workshop** immediately after **Networking Lunch** causes post-meal drowsiness
- **Poster Viewing** immediately after **Networking Lunch** leaves too little time to browse
- **Panel Discussion** before **Keynote** early in the day loses framing context
- **Coffee Break** as the very first session of the day wastes prime morning focus

B. Must NOT violate (hard constraints)

- Must include **Keynote** and **Workshop**
- **Networking Lunch** cannot be the last session of the day

C. Activity Priority

- High-priority: **Keynote, Workshop**
- Medium-priority: **Panel Discussion**
- Low-priority: **Poster Viewing, Networking Lunch, Coffee Break**

- Explain the encoding scheme for this problem and create four parent chromosomes. [4]
- Suggest a suitable fitness function (may be described in words rather than as a formula). Evaluate the fitness of any two chromosomes from part (a). [4]

Solution:

Encoding: Let, Keynote: K, Workshop: W, Poster Viewing: PV, Networking Lunch: NL, Panel Discussion: PD and Coffee Break: CB. Then each chromosome is a set of FIVE sessions chosen from the available sessions. Each chromosome must maintain the hard constraints. [Note: only hard constraints need to be followed / maintained during encoding; soft constraints and other aspects such as priority should be addressed in fitness function, not during encoding. You will be penalized if you do not follow this.]

Four parent chromosomes:

P1 = CB, NL, PV, K, W

P2 = PD, W, PV, K, CB

P3 = K, W, PV, NL, CB

P4 = W, PD, K, NL, PV

Fitness function:

- For each soft constraint violation, a chromosome will get a penalty of 30. (You can choose other values)
- For maintaining each priority, a chromosome will be rewarded according to the priority order. High: 15, Medium: 10, Low: 5. (You can choose other values)

Fitness calculation:

- P1: violates soft constraints 2 and 4, maintains two high priorities and three low priorities. So $f(P1) = -30 * 2 + 15 * 2 + 5 * 3 = -15$
- P2: violates soft constraints 3, maintains two high priorities, one medium and two low priorities. So $f(P1) = -30 * 1 + 15 * 2 + 10 * 1 + 5 * 2 = 20$
- P3: does not violate any soft constraint, maintains two high priorities and three low priorities. So $f(P1) = 15 * 2 + 5 * 3 = 45$
- P4: violates soft constraints 2 and 3, maintains two high priorities, one medium and two low priorities. So $f(P1) = -30 * 2 + 15 * 2 + 10 * 1 + 5 * 2 = -10$